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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Qart.ai
Selected Domain	Domain 1: Aiversecity — Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 – Quality Education (targets 4.4 and 4.3), with SDG 8 – Decent Work and Economic Growth (targets 8.2 and 8.5)
Project / Solution Name	Qart.ai
Name (Leader)	Kydyrgaliyev Meiirman
Email & Phone (Leader)	meirman02.07@gmail.com · +60 11-2725 6756
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	2 (Kydyrgaliyev Meiirman, leader; Mounir Ben Hassine)

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's Global Business Services (GBS) sector employs about 250,000 people, and TalentCorp expects it to reach 276,000 by 2029. Finance and accounting is the core of that work: invoices, cash application, reconciliations and reporting. TalentCorp's 2024 impact study rates 32% of the GBS workforce, about 89,000 people, as highly impacted by AI, digital and green change, and a further 59% as medium impacted.

The evidence is scattered. DOSM shows clerical support work falling from 14.5% to 13.1% of Malaysian employees between 2011 and 2025, while professional roles rose from 13.9% to 18.7%. The Anthropic Economic Index shows AI already being used on the exact tasks these clerks do. The ILO rates clerical work among the most exposed to generative AI. Yet a cash application analyst in Kuala Lumpur, or a graduate about to enter GBS, has no single place that turns this into a personal answer: which of my skills are losing value, which roles can I realistically move to, and what should I learn first, for free, in the hours I have?

Most career tools either list generic jobs or ask a chatbot, which can invent jobs, courses and statistics. Getting it wrong means staying in a role being automated, or training that leads nowhere. This affects decent work and lifelong learning (SDG 8.5, 4.4) for a large, measurable group of Malaysian workers and the graduates who will replace them.

Sources: TalentCorp GBS study 2024, pp. 48, 51; OpenDOSM 2011–2025; Anthropic Economic Index 2026-06-26; ILO WP140 (2025).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Qart.ai takes a person from "who I am today" to "where I can be in the AI era". A user uploads a synthetic CV or answers six short questions. Personal details are removed first. OpenAI gpt-4o-mini then reads the CV into skill claims, but every claim must quote a sentence from the CV, and code rejects any quote that is not really there.

A scheduled pipeline collects dated public evidence from seven registered sources: OpenDOSM, DOSM job ads, the Anthropic Economic Index, TalentCorp, O*NET, the ILO exposure index and MyCOL. Code, not the language model, turns that evidence into numbers: how much each skill matters under a moderate and a

fast AI-adoption scenario, an explainable fit score for nine real GBS destination roles, and a skills gap. Every number is labelled observed, assumption or estimate, with its source and date.

The plan is shown as three roads: AI basics first, then reskilling without AI and reskilling with AI. Each stop opens checked free courses, YouTube videos and articles. A comparison shows each route's hours and fit in both scenarios. Every role and step is tagged to an SDG target and explains how it contributes. New evidence proposes updates; the person accepts or declines them.

Two open models run on a RunPod serverless GPU: BGE-M3 embeddings and the BGE reranker. They match any job title, in English or Malay, to 1,016 occupations and map free-text tasks to skills. In our test of 40 labelled titles, top-1 accuracy rose from 58% (keyword search) to 72%, and for Malay titles from 0% to 75%.

Workshop skills used: GenAI APIs (schema-constrained gpt-4o-mini), database architecture (one SQLAlchemy layer, PostgreSQL on Replit), cloud hosting (Replit Autoscale and a Scheduled deployment) and GPU inference on RunPod.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Built and deployed (live at https://qartai.replit.app/start):

Frontend: React 19 + TypeScript, Vite, Tailwind CSS, Recharts (charts), react-router.

Backend: Python 3.12, FastAPI, Pydantic, Uvicorn.

Database: SQLAlchemy Core over SQLite (local) and PostgreSQL (Replit production database).

GenAI API: OpenAI gpt-4o-mini with strict JSON schemas; every answer validated in code; labelled fallbacks.

GPU models: BAAI/bge-m3 (embeddings) and BAAI/bge-reranker-v2-m3 on a RunPod serverless endpoint (queue-based, scales to zero).

Data pipeline: 7 collectors (OpenDOSM, DOSM Job Market Insights, Anthropic Economic Index, TalentCorp, O*NET 31.0, ILO WP140, MyCOL 2024/25), an immutable evidence store, backtests.

Hosting: Replit Autoscale (web app and API) + Replit Scheduled deployment (daily collection), one shared database.

Quality: 175 automated tests (pytest), ruff, mypy, oxlint; fairness test with paired synthetic CVs.

System block diagram: Appendix A at the end of this document.

Sponsor-provided GenAI API or Cloud Platform: Yes. Replit hosts the deployed app (Autoscale), its PostgreSQL production database, and a Scheduled deployment that runs the evidence pipeline daily without anyone visiting. RunPod serverless GPU serves the two open models (BGE-M3, BGE reranker) for occupation and skill matching, with a time budget and a labelled keyword fallback if the GPU is cold or unavailable. Both were used with sponsor-provided credits. The GenAI API (OpenAI gpt-4o-mini) runs on the team's own paid key.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define: we started from the people in the brief and narrowed to one measurable group: GBS finance and accounting staff in roles TalentCorp rates highly impacted, and the graduates entering them. Their moment of need: a restructuring notice or a first job search, asking what to learn next.

Ideate: we rejected "upload a CV and ask a chatbot" because it invents jobs and courses. Instead, code computes every number from dated evidence, and the language model only reads CVs and rephrases computed facts.

Prototype and test: we built in small slices and tested each against the real API: 175 automated tests, a fairness test where CVs differing only in name, gender, religion, race and age give identical results, and a measured RunPod evaluation. We walked through the app as users, collected feedback and redesigned the weakest screens: the roadmap, the plan comparison, the CV reading log and crowded layouts.

Deploy: the app is live on Replit with a production database and a daily scheduled data run. Every remote service has a time budget and a labelled fallback, so no page waits on a cold GPU. For Phase 2 we will test with real users and widen coverage.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Supported by Target 8.5 (full and productive employment and decent work for all) and 8.2 (productivity through technological upgrading and innovation).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	32% of Malaysia's GBS workforce (about 89,000 people) work in roles rated highly impacted by AI, digital and green change (TalentCorp 2024, PDF p. 48). Today these workers have no single tool that links this evidence to a personal plan. In our prototype's verified learning catalogue, 27 of the 37 skills (73%) the destination roles need have a verified free course today; the rest have only videos or articles. For our three synthetic test profiles, the average fit for their best destination role starts at 11.7 / 100 (moderate scenario).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	(1) Coverage: % of required skills with a verified free course. (2) Projected readiness (estimate): average fit gain for the synthetic test profiles after following the recommended route, computed by the same engine. (3) Time to a plan: share of test users who get from opening the app to ranked roles and a pathway in under 5 minutes. (4) Traceability: % of numbers shown with source and date (target 100%, checked by tests).

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	(1) Raise verified free-course coverage from 73% to at least 90% of required skills. (2) Keep the projected fit gain on the with-AI route at 50 points or more on average (today: 11.7 → 77.5, estimate), while showing the without-AI route honestly (today: 11.7 → 38.8). (3) At least 80% of at least 10 test users reach a plan in under 5 minutes (compute time today: 2–5 seconds from CV to plan). (4) 100% of numbers keep their source and date. Sources: TalentCorp (2024) GBS impact study, PDF p. 48; UN SDG target text from sdgs.un.org; fit and coverage measured in the Qart.ai prototype on 24 October 2026 (synthetic profiles "Nur", "Hafiz" and "Mei"); projected fit is an estimate from declared course rules, not a measured learning outcome.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI, Pydantic, Uvicorn, SQLAlchemy, httpx, PyYAML	code	MIT / BSD-3-Clause
psycopg 3 (PostgreSQL driver)	code	LGPL-3.0 (used unmodified)
pypdf, python-docx, python-multipart	code	BSD-3-Clause / MIT / Apache-2.0
openai Python SDK	code	Apache-2.0
React, react-router, Recharts, lucide-react, Vite, Tailwind CSS, TypeScript	code	MIT / ISC / Apache-2.0
sentence-transformers, transformers, PyTorch (RunPod worker only)	code	Apache-2.0 / BSD-3-Clause

BAAI/bge-m3	model	MIT
BAAI/bge-reranker-v2-m3	model	Apache-2.0
OpenAI gpt-4o-mini	model (API)	OpenAI API terms
O*NET 31.0 Database	dataset	CC BY 4.0
Anthropic Economic Index (release 2026-06-26)	dataset	CC BY
ILO WP140 GenAI exposure scores	dataset	CC BY 4.0
OpenDOSM / DOSM Job Market Insights	dataset	Malaysian Government Open Data Terms 1.0
TalentCorp GBS impact study (2024), MyCOL 2024/25	dataset (facts with page citations)	All rights reserved; facts cited, no text reproduced
ESCO–O*NET crosswalk	dataset	ESCO terms (European Commission)
SDC Hackathon submission template (organisers)	code (repository starting point)	Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

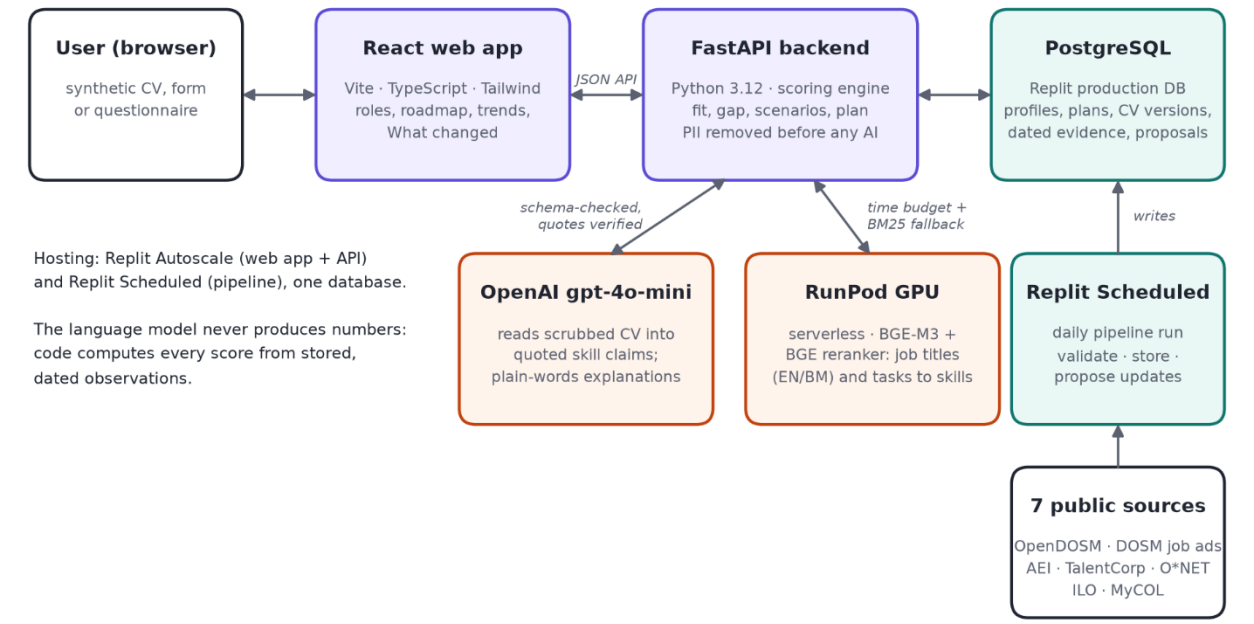
7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Kydyrgaliyev Meiirman, Team Leader

25 October 2026

Appendix A: System block diagram



How the parts connect: the React web app and FastAPI backend run on Replit Autoscale; OpenAI gpt-4o-mini and the RunPod GPU models are called with time budgets and labelled fallbacks; a Replit Scheduled deployment collects the seven public sources daily into the shared PostgreSQL database.